

Skills

Languages: Python, R, SQL, TypeScript, JavaScript

ML/Data: PyTorch, OpenCV, Roboflow, SAM2, PaddleOCR, scikit-learn, Pandas, NumPy, SciPy, statistical modeling, Elo systems, Monte Carlo simulation

Apps/Infra: R Shiny, RMarkdown, FastAPI, Next.js, React, PostgreSQL, Supabase, Drizzle ORM, Streamlit, Recharts, Plotly, Grafana, Power BI, Git

AI Engineering: LLM application design, tool/function calling, MCP concepts, RAG patterns, agent workflows, structured outputs, model evaluation

Experience

STARS Advanced Radiology Services

Grand Rapids, Michigan (Remote)

Data Analyst

04/2022 - Present

- Build production R Shiny apps and code-free SQL query tools for operations, billing, and leadership teams, turning stakeholder requests into validated dashboards, recurring analyses, and troubleshooting workflows.
- Automated RMarkdown worklist prediction and patient wait-time reporting, helping optimize radiologist workflows and reduce unread worklists by 60%.
- Lead SQL-based billing and payer reporting, surfacing trends that contributed to a 40% reduction in Days A/R and an 85% reduction in aged A/R dollars over 120 days.

Byrne Electrical Specialists Inc.

Rockford, Michigan

Data Science Intern

(Summers 2019 - 2021)

- Built PostgreSQL-backed IoT sensor pipelines, Grafana/Power BI dashboards, and SPC alert logic for robotic assembly OEE, contributing to a 15% improvement in machine OEE.

Selected Projects - live demos at bendotwillcox.github.io/portfolio

ValoMapped - Esports Elo Analytics Platform | Next.js, TypeScript, PostgreSQL, Drizzle ORM, Recharts

- Built and deployed a live Valorant Champions Tour analytics platform with automated ingestion, map-specific Elo ratings, match predictions, player ratings, record books, pick/ban analysis, grid-searched rating calibration, and Monte Carlo tournament simulations.

HoopStats - AI Basketball Video Analytics Pipeline | Python, PyTorch, OpenCV, Roboflow, SAM2

- Implemented video I/O, RF-DETR detection, ByteTrack/SAM2-style tracking, team color classification, court homography, trajectory normalization, and early CSV export scaffolding for a broadcast basketball analytics pipeline.

Voting Paradigm - Voting Systems and Social Choice Demos | Python, FastAPI, Next.js, TypeScript, Supabase

- Built a hosted social-choice demo portfolio with voting-method comparison, apportionment/districting, preference-agent replay, and a tested Python package for plurality, approval, IRV, Borda, Ranked Pairs, score, and quadratic voting.

Education

Michigan State University - B.S. Economics, Minor in Quantitative Data Analytics | Dec 2021